

Does Sheaf Cohomology Add Real Power to Federated Witness-Log Systems? A Literature Review, Assessment, and Prototyping Plan

TL;DR

- **Yes, conditionally.** Sheaf *cohomology* adds genuine, non-redundant power over pairwise digest comparison in exactly one regime: **partial visibility** (partitions, incomplete gossip, asynchronous exchange) where no global comparison is possible but local cocycle conditions over overlapping triangles still bind — structurally identical to Abramsky–Brandenburger contextuality, where locally consistent data has no global section. In the *full-visibility* regime, H^1 genuinely reduces to renamed equivocation-detection, and the volume should say so honestly.
- The **operational payoff** is quantitative and localizing: over a real-vector-space embedding of log/delta states, $\dim H^1$ (equivalently, the dimension of the sheaf-Laplacian kernel on edges) counts independent equivocation classes; the support of the minimal-norm harmonic 1-cochain localizes equivocators to a subgraph; and Robinson’s consistency radius gives a computable divergence magnitude — all in polynomial time via linear algebra (the #P-hardness of sheaf cohomology applies only to coherent sheaves on projective space, not finite cellular sheaves over $\mathbb{R}$).
- **Commit-or-cut verdict:** COMMIT the chapter if a prototype shows H^1 /spectral signals detect and localize equivocators under partition where forensic pairwise methods provably cannot — the Sheng et al. (CCS’21) impossibility, verbatim, is that “all secure BFT protocols designed for $2t+1$ replicas communicating over a synchronous network forensic support is inherently nonexistent; this impossibility result holds for all BFT protocols and even if one has access to the states of all replicas (including Byzantine ones).” [Semantic Scholar](#) CUT it to an appendix remark if every equivocation your experiments detect is equally detectable by $O(|E|)$ digest comparison — the honest falsifying outcome.

Key Findings

1. The toy sheaf result is real but weak; cohomology is where the question lives.

That prefix-consistent logs over a gossip graph form a sheaf (pairwise-compatible local

sections glue uniquely iff no equivocation) is precisely the statement that $H^0 =$ global sections and that the gluing axiom holds — the degree-0 story. Whether H^1 and the sheaf Laplacian spectrum add power is genuinely open and maps cleanly onto three mature literatures.

2. Equivocation is contextuality. The deepest structural match is Abramsky & Brandenburger, “The Sheaf-Theoretic Structure of Non-Locality and Contextuality” (*New J. Phys.* 13:113036, 2011; arXiv:1102.0264) and its cohomological sequel Abramsky, Mansfield & Barbosa, “The Cohomology of Non-Locality and Contextuality” (*EPTCS* 95:1–14, 2012; arXiv:1111.3620). The 2011 paper states verbatim: “We show that contextuality, and non-locality as a special case, correspond exactly to obstructions to the existence of global sections. We describe a linear algebraic approach to computing these obstructions.” [arXiv](#) Equivocation is exactly this — each peer’s local view is internally consistent and consistent with neighbors it has compared with, yet no global witness history exists. This is the single most important analogy for the volume.

3. The computational toolkit is mature and directly reusable. Hansen & Ghrist’s “Toward a Spectral Theory of Cellular Sheaves” (*J. Appl. Comput. Topology* 3(4):315–358, 2019; arXiv:1808.01513) provides the sheaf Laplacian $L_0 = \delta^\top \delta$ and up-Laplacian $L_1 = \delta \delta^\top$, with Hodge isomorphism $\ker L_k \cong H^k(\text{jakobhansen } G; F)$ over $\mathbb{R}$. Curry’s thesis “Sheaves, Cosheaves and Applications” (UPenn 2014; arXiv:1303.3255) is the foundational reference for cellular sheaves; [arXiv](#) Curry–Ghrist–Nanda “Discrete Morse Theory for Computing Cellular Sheaf Cohomology” (*Found. Comput. Math.* 16(4):875–897, 2016) gives efficient computation. Robinson’s *Topological Signal Processing* (Springer, 2014) and consistency-radius work supply a metric (non-cohomological) measure of divergence.

4. Almost nobody has applied this to Byzantine/consensus systems specifically — a genuine gap the volume can fill. Searches for sheaf + consensus/blockchain/Byzantine return mostly sensor-fusion/opinion-dynamics literature repurposed, plus a small number of speculative pieces. The state-of-the-art non-sheaf baseline is Sheng, Wang, Nayak, Kannan & Viswanath, “BFT Protocol Forensics” (CCS’21, pp. 1722–1743). [princeton](#) [dblp](#) The comparison against this baseline — not against toy examples — is what will make or break the chapter’s credibility.

Details

PART A — Literature Review

A.1 Sheaves in distributed systems and concurrency (the 1990s lineage).

- **Joseph Goguen, "Sheaf Semantics for Concurrent Interacting Objects," *Math. Structures in Comp. Sci.* 2(2):159–191, 1992.** Objects modeled as sheaves over a poset of time intervals; systems as diagrams; behavior as limit; interconnection as colimit. [Cambridge Core](#) Explicitly treats deadlock, non-interference (security), inheritance. [Amazonaws](#) The origin point for "sheaf = local behavior gluing to global behavior" in CS. Relevance: it establishes the sheaf-condition-over-time formalism but does *not* use cohomology — exactly the gap this program probes.
- **Monteiro & Pereira** sheaf-theoretic object interconnection (within the Goguen lineage); **Lilius, "Sheaf Semantics for Petri Nets"** (Helsinki Univ. of Technology, Digital Systems Lab, 1993) extends the semantics to Petri nets.
- **Joyal, Nielsen & Winskel, "Bisimulation from Open Maps"** (*Information and Computation* 127(2):164–185, 1996) and the **presheaf models of concurrency** program (Cattani & Winskel, "Presheaf Models for Concurrency," BRICS/CONCUR; Cattani's 1999 BRICS thesis DS-99-1). Here presheaves on a "path category" (pomsets) model processes, with bisimulation characterized via open maps and spans. [arxiv](#) A *presheaf* (not cohomological) use — relevant because it shows the category-theoretic machinery for "local observations glue to global process," and because the "path category" idea is a candidate site for log states.
- **Assessment:** This lineage gives semantic foundations but is orthogonal to quantitative invariants. It supports modeling choices (poset site, presheaf of local states) but offers no counting/localization tools. Cite it for pedigree, not for power.

A.2 Applied/cellular sheaf theory and spectral methods (the toolkit).

- **Curry (2014)** — cellular sheaves of vector spaces on cell complexes; the computable backbone. [arXiv](#) [University of Pennsylvania Mathen](#) **Hansen & Ghrist (2019)** — sheaf Laplacian, harmonic cochains, eigenvalue interlacing, effective resistance, Cheeger inequality, sheaf approximation. [Springer](#) **Hansen & Ghrist, "Opinion Dynamics on Discourse Sheaves"** (*SIAM J. Appl. Math.* 81(5):2033–2060, 2021; DOI 10.1137/20M1341088) — models lying/deception/propaganda as sheaf structure, diffusion via the sheaf Laplacian; [ResearchGate](#) directly analogous to modeling a dishonest harbor. **Hansen & Ghrist, "Distributed Optimization with Sheaf Homological Constraints"** (Allerton 2019) — H^1 constraints in distributed optimization.
- **Robinson, *Topological Signal Processing*** (Springer, 2014); "Sheaves are the canonical data structure for sensor integration" (*Information Fusion*, 2017); "Assignments to sheaves of pseudometric spaces" (*Compositionality*, [arXiv:1805.08927](#), 2020). The **consistency radius**: for an assignment a to a sheaf S of pseudometric spaces on poset $(P, \leq)$, $c_S(a) = \sqrt{(\sum_{x \leq y} d_y(a_y, F(x \leq y)(a_x))^2)}$;

[arxiv](#) it lower-bounds distance to the nearest global section (Prop. 2.3: if every restriction has Lipschitz constant $\leq K$, then $c_S(a) \leq (1+K) \cdot d_S(s,a)$ for every global section s), [arxiv](#) and thresholding yields the *consistency filtration* (a functor, robust to perturbation). [arXiv](#) This is the metric analogue of “how badly is the system equivocating.”

- **PySheaf** (github.com/kb1dds/pysheaf, Michael Robinson, Brenda Praggastis, Janelle Henrich) — Python 3.6 cellular-sheaf library computing consistency radius and fuseAssignment [GitHub](#) (data fusion via consistency-radius minimization). **Critical limitation for this program:** the maintainers’ own release notes state verbatim: “Cohomology is not presently supported. While the cochain complex exists mathematically for sheaves of abelian groups on partial orders, it is not necessarily efficient to compute! We anticipate adding this capability in the future.” [GitHub](#) Dependencies: scipy, numpy, networkx, deap. [GitHub](#) Implication: for H^1 you will hand-roll coboundary matrices in numpy/scipy; PySheaf is usable only for the consistency-radius track.

A.3 Contextuality / Čech cohomology of “locally consistent, globally inconsistent” data (the structural heart).

- **Abramsky–Brandenburger (2011):** contextuality/non-locality $\iff$ no global section of a presheaf of measurement outcomes; a linear-algebraic method computes the obstruction; a strict hierarchy (Bell < Hardy < GHZ) of no-go strengths. [arXiv](#)
- **Abramsky–Mansfield–Barbosa (2011/2012):** define a Čech cohomology obstruction on an *abelian* presheaf $F_R \circ S$ (free R -module on the support), giving a cohomology class that **vanishes if a global section exists**; non-vanishing is a **sufficient but not necessary** condition [ADS](#) for contextuality (abstract, verbatim: “the non-vanishing of the obstruction provides a sufficient (but not necessary) condition for the model to be contextual”). [arXiv](#) [ADS](#) Verified examples: PR boxes, GHZ, Peres–Mermin square, 18-vector Kochen–Specker. [arXiv](#)
[University of Edinburgh Research](#)
- **Honest caveat (false negatives):** the obstruction is sufficient, not necessary. AMB themselves note the Hardy model escapes detection. **Giovanni Carù (“On the Cohomology of Contextuality,” arXiv:1701.00656, 2017; “Towards a complete cohomological invariant...,” arXiv:1807.04203, 2018)** exhibits a strongly contextual model that is *not* cohomologically contextual even under symmetry/connectedness, disproving Conjecture 8.1 of AMB — i.e., Čech cohomology “is not an invariant for strong contextuality, and it can fail even on extremely simple scenarios.” [University of Oxford](#) **Abramsky, Barbosa, Kishida, Lal & Mansfield, “Contextuality, Cohomology and Paradox” (CSL 2015, LIPIcs**

41:211–228; arXiv:1502.03097) shows Čech cohomology *is* complete for the “All-vs-Nothing” [arXiv](#) class — so completeness holds only in special classes. This transfers directly: a vanishing H^1 over $\mathbb{R}$ certifies consistency only up to the abelianization gap; genuine set-valued/non-linear inconsistency can be cohomologically invisible.

A.4 2020–2026 work on sheaves in consensus/ledgers/faults (thin, mostly adjacent).

- Non-sheaf baseline (the one to beat): Sheng et al., “BFT Protocol Forensics” (CCS’21, pp. 1722–1743; arXiv:2010.06785).** Formalizes *forensic support*, verbatim from the abstract: “we aim to identify (with cryptographic integrity) as many of the malicious replicas as possible and in as distributed manner as possible. [ACM Digital Library](#) Our main result is that forensic support of BFT protocols depends heavily on minor implementation details that do not affect the protocol’s security or complexity.” [Semantic Scholar](#) The key impossibility (located in Appendix B of the paper; the five protocols studied are PBFT, HotStuff, Algorand, LibraBFT, and VABA) [NSF PAGES](#) reads verbatim: “all secure BFT protocols designed for $2t+1$ replicas communicating over a synchronous network forensic support is inherently nonexistent; this impossibility result holds for all BFT protocols and even if one has access to the states of all replicas (including Byzantine ones).” Follow-ups: “CFT-Forensics” (AFT 2024); Diem/DiemForensics. Equivocation (a signer publishing incompatible signed roots) is exactly the evidence forensics exploits — a signed pair of conflicting roots is a non-repudiable culprit proof.
- Key transparency / gossip auditing (the direct application domain): CONIKS** (USENIX Security 2015) — signed chained directory snapshots; equivocation to two parties “must be maintained forever or else it will be detected by auditors who can then broadcast non-repudiable cryptographic evidence.” **“Think Global, Act Local: Gossip and Client Audits in Verifiable Data Structures”** (arXiv:2011.04551) and **EthIKS** — gossip/blockchain to prevent split-view/equivocation. [ResearchGate](#) **MINGLE** (IACR eprint 2026/1010) — in-band gossip piggybacking KT commitments [IACR](#) to detect split-view equivocation *without* dedicated auditors: “whether clients can detect equivocation themselves, without dedicated infrastructure, simply by comparing KT state as they communicate.” [IACR](#) MINGLE is the closest existing system to the volume’s model and the natural real-world anchor.
- CRDTs / semilattices (the coefficient-structure problem): state-based CRDTs** require the state space to be a **join-semilattice** with commutative/associative/idempotent merge = least upper bound [arxiv](#) (Shapiro et al.). Prefix-ordered logs are exactly such a semilattice. This is the crux obstruction for cohomology: **Čech cohomology beyond H^0/H^1 needs abelian-group**

coefficients, and a semilattice is not a group. Delta-CRDTs ("Efficient State-based CRDTs by Delta-Mutation," arXiv:1410.2803) motivate the delta-vector abelianization route.

- **Speculative/adjacent:** "The Sheaf Laplacian: A Topological Framework for Data Fusion and Consensus in Distributed Sensing Networks" (arXiv:2606.19529, 2026) — anomaly = violation of the network's relational contract, detected via Dirichlet-energy spikes, faulty node located by energy contribution; [arXiv](#) the sensor-network cousin of equivocation localization but *not* adversarial/cryptographic. "Blockchain Cohomology" (Meldman-Floch, CEUR Vol-2478) and a 2025 "Sheaf Cohomology for Fragment-Sequencing in Hierarchical Block Rectangular Matrices" (Scifiniti) exist but are peripheral and of uneven rigor — cite with caution, not as load-bearing.
- **ML tooling (reusable code only): Bodnar et al., "Neural Sheaf Diffusion"** (NeurIPS 2022; arXiv:2202.04579) with public code (github.com/twitter-research/neural-sheaf-diffusion) — reusable PyTorch machinery for building coboundary maps and sheaf Laplacians over graphs and *learning* restriction maps from data [arXiv](#) (relevant if restriction maps are not hand-specified). Barbero et al. "Sheaf Neural Networks with Connection Laplacians" (2022). Hanks, Riess et al. "Distributed Multi-Agent Coordination over Cellular Sheaves" (arXiv:2504.02049, 2025) — Sheaf-ADMM, the closest to a distributed-algorithms use.

PART B — Assessment (the intellectual core)

B.0 The coefficient problem, stated precisely. Prefix-ordered log states form a join-semilattice $(S, \sqcup)$, not an abelian group. Čech/cellular cohomology in degree ≥ 1 needs abelian-group (or R-module) coefficients so that the coboundary δ and the cocycle condition $\delta^2=0$ make sense with subtraction. Three escape routes, of which the assessment must commit to one (or two, by track):

- **(a) Abelianization via delta/balance vectors (RECOMMENDED for the economic ledger).** For the balance-vector ledger, each harbor's state *is* already a vector in $\mathbb{Z}^k$ or $\mathbb{R}^k$ (balances of k accounts); differences are well-defined; this is a genuine abelian group and full cohomology applies. For the append-only log, embed each log state by its **delta vector**: the multiset/vector of new entries since a common ancestor (cf. delta-CRDTs). Two prefix-compatible states have delta vectors that agree on their overlap; equivocation = delta vectors that cannot be simultaneously realized. The cleanest path to a real H^1 .
- **(b) Nonabelian H^1 as torsors/gluing data.** For the pure semilattice, the correct object is the *pointed set* $H^1(G; F)$ classifying F -torsors — gluing data for local sections that may fail to glue. The honest home for equivocation-as-obstruction

without forcing a group structure, but it yields a set, not a dimension: you get “is there an obstruction / how many gluing classes,” not a spectrum. Good for existence theorems, poor for quantitative counting.

- **(c) Hansen–Ghrist real relaxation (RECOMMENDED for algorithms/localization).** Embed log states as vectors in $\mathbb{R}^d$ (e.g., Merkle-root hashes mapped to $\mathbb{R}^d$, or delta vectors), restriction maps as linear maps (projections onto shared-prefix coordinates), and measure equivocation as a **nonzero harmonic residual** — the minimal-norm element of $\ker L_1 \cong H^1(G;F)$. This sacrifices exactness (route (a)/(b)’s clean iff) for a robust, spectral, localizable, noise-tolerant signal, and is the route where the Laplacian toolkit and PySheaf-style optimization pay off.

B.1 The right site and the sheaf that makes H^1 nontrivial exactly under equivocation.

- **Base space:** the gossip graph G itself as a 1-dimensional cell complex (vertices = harbors, edges = peer links actually used for exchange). For cocycle conditions in degree 2 you need 2-cells: take the **nerve of the peer-overlap cover** — fill a triangle $\{u,v,w\}$ with a 2-cell iff all three pairwise-exchanged and their views share a common comparison point. This is the exact analogue of the measurement cover in Abramsky–Brandenburger; the triangles are where the cocycle condition binds under partial visibility.
- **Stalks:** $F(v)$ = the (embedded) local log/delta state at v ; $F(e)$ = the shared-prefix state that u and v have compared on edge $e = \{u,v\}$. **Restriction maps:** $F(v) \rightarrow F(e) =$ “restrict my state to the prefix I showed peer along e .”
- **H^0 = global sections = a single consistent witness history** (the toy result; Hansen–Ghrist confirm $H^0(X;F)$ is naturally isomorphic to the space of global sections). [arxiv](#) **$H^1 \neq 0$ exactly when there exist locally compatible restrictions with no global gluing** = equivocation *under the visibility structure encoded by the cover*. Crucially, in a fully-connected comparison (every pair compares every root), the cover is trivial and H^1 collapses to pairwise-mismatch detection. In a *partial* cover (partitions, sparse gossip), H^1 can be nonzero even when no single pair has caught a mismatch — this is the value-add.

B.2 What H^1 rank / Laplacian spectrum counts operationally. Using the graph two-term complex $0 \rightarrow C^0 = \bigoplus_v F(v) \xrightarrow{\delta} C^1 = \bigoplus_e F(e) \rightarrow 0$:

- $\dim H^0 - \dim H^1 = \sum_v \dim F(v) - \sum_e \dim F(e)$ (the sheaf Euler characteristic — Hansen–Ghrist 2019; the standard alternating-sum identity of a finite two-term cochain complex).
- $\dim H^0 = \dim \ker \delta = \dim \ker L_0 =$ number of independent globally-consistent

assignments (Hodge isomorphism $\ker L_k \cong H^k$; Hansen–Ghrist §2.2.2, §3).

- **$\dim H^1 = \dim C^1 - \text{rank } \delta = \dim \ker L_1$** = the number of **independent equivocation classes** — independent ways local views fail to reconcile. Each independent conflicting-root pair that cannot be routed around a cycle contributes to H^1 .
- **Localization:** the **support of the minimal-norm harmonic 1-cochain** (minimizer of $\|\cdot\|$ in $\ker L_1$) concentrates on the edges/cycles carrying the inconsistency, localizing equivocators to a subgraph — the exact mechanism the sensor-network paper (arXiv:2606.19529) uses for faulty-node location via Dirichlet-energy contribution. [arXiv](#)
- **Repair:** the minimum edge set whose removal kills H^1 is a **cohomological min-cut**; Ghrist–Krishnan, “A Topological Max-Flow-Min-Cut Theorem” (GlobalSIP 2013, pp. 815–818, DOI 10.1109/GlobalSIP.2013.6737016) encode capacity constraints “as a sheaf of semimodules over the network” with “flow and cut values as a directed homology taking values in the sheaf,” [IEEE Xplore](#) and Ghrist–Hiraoka network-coding sheaves (NOLTA 2011) are the relevant precedents — with the important caveat that the 2013 duality has known gaps (tightened in Krishnan’s later “Flow-Cut Dualities for Sheaves on Graphs” and Krishnamoorthy et al., *J. Comb. Optim.* 2024, arXiv:2107.04252). [arXiv](#)

B.3 Comparison against non-sheaf baselines.

- **Full visibility:** pairwise digest comparison is $O(|E|)$ and detects any equivocation a single edge witnesses; forensic BFT (Sheng et al.) additionally produces cryptographic culprit proofs. Here cohomology is **strictly redundant** — $H^1 \neq 0$ iff some edge sees a mismatch. The chapter must concede this.
- **Partial visibility / partitions (the win):** when pairwise comparison is impossible (peers never directly compared, or a partition heals only transitively), the **cocycle condition over triangles still binds**: three peers who each pairwise-agree on overlaps can be forced into global inconsistency detectable only cohomologically. This is the precise transfer of Abramsky–Brandenburger’s “locally consistent, globally inconsistent.” Sheng et al.’s impossibility (no forensic support for $2t+1$ synchronous protocols even with all replica states) marks exactly where pairwise/forensic methods provably stop; the open empirical question is whether cohomology extends detection into that gap.
- **Robustness/noise:** the consistency radius (Robinson) and Laplacian spectrum degrade gracefully under noisy/approximate matching, where exact digest equality is brittle. Under Byzantine reordering or lossy gossip, a spectral signal may survive where a single-bit digest comparison fails.

B.4 Three theorem candidates (strongest first) with proof-strategy sketches.

Theorem 1 (Counting — the anchor). *Let G be a gossip graph with the peer-overlap cover, and F the delta-vector coefficient sheaf (route a) with $F(v)=\mathbb{R}^{\{d_v\}}$, $F(e)=\mathbb{R}^{\{d_e\}}$ and restriction maps the shared-prefix projections. Then $\dim H^1(G;F) \geq (\text{number of independent equivocation classes})$, with equality iff the honest sub-sheaf is flabby (every locally consistent family over the honest subgraph extends). Proof strategy: Build δ explicitly; each equivocation = a pair of restrictions to a shared edge whose images differ, i.e., a class in $\text{coker } \delta = H^1$. Independent equivocations give linearly independent coker representatives $\Rightarrow \geq$. For equality, flabbiness/acyclicity of the honest sub-sheaf kills spurious H^1 from topology alone (constant-sheaf H^1 of G = cycle rank β_1), isolating H^1 to genuine equivocation. Use the Euler-characteristic identity to separate topological H^1 (cycles) from coefficient H^1 (equivocation) — a subtlety the chapter must handle: on a cyclic gossip graph even the constant sheaf has $H^1 = \mathbb{R}^{\{\beta_1\}}$, so equivocation H^1 must be measured *relative to the honest baseline*.*

Theorem 2 (Localization). *The support of the minimal- ℓ^2 harmonic 1-cochain $h \in \ker L_1$ representing an equivocation class is contained in the union of cycles passing through the equivocating signer(s); hence equivocators are localized to within graph-distance r of $\text{supp}(h)$, where r is bounded by the diameter of the smallest cover-triangulated region carrying nonzero h . Proof strategy: Harmonic $\Rightarrow h$ minimizes Dirichlet energy $h^T L_1 h$; energy concentrates on edges with inconsistent restrictions; a discrete unique-continuation / effective-resistance argument (Hansen–Ghrist §6) bounds spread. Adapt the faulty-node-location argument of arXiv:2606.19529 to the adversarial signed setting.*

Theorem 3 (Consistency radius = max prefix divergence, computable). *For the log presheaf with stalks under the prefix pseudometric, the Robinson consistency radius $c_S(a)$ equals the maximum over compared peer-pairs of their prefix-divergence (length of the disagreeing suffix), computable in $O(|E| \cdot L)$ time where L bounds compared-log length; $c_S(a)=0$ iff no equivocation is witnessed by the cover. Proof strategy: Direct from the consistency-radius definition (Robinson, arXiv:1805.08927, Def. 2.2) [arxiv](#) specialized to the prefix pseudometric; the sum/sup over pairs is a max over edges; the lower-bound-to-nearest-section proposition (Lipschitz constant K of prefix projection = 1) gives $c_S(a) \leq 2 \cdot d(\text{section}, a)$. This is the “cheap, honest, metric” result that stands even if Theorems 1–2 fail.*

B.5 Honest limits. (i) The abelianization gap (Carù counterexamples): vanishing H^1 over $\mathbb{R}$ need not certify true consistency for set-valued log states. (ii) On cyclic gossip graphs, topological H^1 (cycle rank β_1) contaminates the equivocation signal and must be quotiented out. (iii) Cryptographic non-repudiation (the forensic-BFT deliverable) is *not* produced by cohomology — H^1 says “inconsistency exists here,” not “signer X signed

these two roots"; the two approaches are complementary, and the chapter should frame cohomology as *triage/localization* feeding forensic proof extraction.

PART C — Prototyping Plan (headless Linux container)

C.1 Stack. Python 3.11, numpy + scipy (sparse linalg: `scipy.sparse`, `scipy.sparse.linalg.eigsh` for Laplacian spectra, `numpy.linalg.matrix_rank` /SVD for δ rank), networkx for graph generation, optionally the neural-sheaf-diffusion repo's coboundary utilities for learned restriction maps. **PySheaf** only for the consistency-radius/fuseAssignment track (its maintainers confirm it cannot compute cohomology). Everything else hand-rolled — deliberately small linear algebra.

C.2 Data structures. Gossip graph G in networkx. Orient edges arbitrarily for signs. Stalks: dict $v \mapsto \mathbb{R}^{d_v}$ (embedded log/delta state), $e \mapsto \mathbb{R}^{d_e}$. Restriction maps: dict $(v,e) \mapsto$ dense/sparse matrix. Assemble the block coboundary δ ($|C^1| \times |C^0|$) with signed blocks $[\sigma:\tau] \cdot F_{\{v \preceq e\}}$. Compute: $\text{rank}(\delta) \rightarrow \dim H^0 = |C^0| - \text{rank}$, $\dim H^1 = |C^1| - \text{rank}$; $L_0 = \delta^T \delta$, $L_1 = \delta \delta^T$; harmonic residual = projection onto $\ker L_1$ via smallest eigenpairs; consistency radius by the closed-form max over edges.

C.3 Experiment matrix.

- **Topologies:** ring (pure cycle — tests β_1 contamination), random 3-regular expander (good mixing — tests fast detection), hub-and-spoke/star (tests localization to hub), two clusters joined by a bridge that is severed then healed (**partition** — the key regime), and Barabási–Albert (realistic federation).
- **Adversary placements:** 0 (honest baseline — must give $H^1 =$ topological β_1 only, consistency radius 0), 1 equivocator (leaf vs hub vs cut-vertex), $k = 2 \dots \lfloor n/3 \rfloor$ equivocators (colluding vs independent), and the **partition-straddling** equivocator showing different roots to the two sides.
- **Coefficient variants:** (a) delta-vectors over $\mathbb{R}$ (exact-ish), (c) random $\mathbb{R}^d$ hash embeddings (robust/relaxed), and a set-valued control to expose the abelianization gap.
- **Metrics:** does $\dim H^1$ (relative to honest baseline) = number of independent equivocations? does $\text{supp}(\text{harmonic})$ contain the equivocator's incident edges? detection recall/precision vs $O(|E|)$ pairwise digest comparison, *especially under partition*; runtime scaling to $n = 10^3$ – 10^4 nodes.

C.4 Baselines to beat. (1) $O(|E|)$ pairwise digest/root comparison. (2) A forensic-style culprit-identification pass (Sheng et al. semantics) on the same traces. (3) Robinson consistency-radius-only (no cohomology). The chapter's claim survives only if cohomology strictly dominates (1)–(3) somewhere — the partition regime is the

predicted somewhere.

C.5 What a falsifying outcome looks like (pre-registered honesty). If, across all topologies and adversary placements, every equivocation flagged by H^1 /spectral signals is *also* flagged by $O(|E|)$ pairwise comparison at equal or lower cost, and the partition regime yields no cohomology-only detections (because healed partitions reduce to eventual pairwise comparison), then **cohomology adds nothing beyond renamed equivocation-detection** and the volume should CUT the chapter to a short “sheaf-theoretic framing” remark. Concretely: if the count-of-detections curve for cohomology never exceeds the pairwise curve, cut.

Recommendations

Stage 1 (2–3 weeks, decisive): Build the hand-rolled numpy/scipy prototype (C.1–C.2). Run the honest baseline and single-equivocator cases on ring + partition topologies with delta-vector coefficients. **Benchmark that flips the decision:** if $\dim H^1$ (net of β_1) does not equal the number of injected equivocations, or if the partition case yields no cohomology-only detection, stop and cut. If both hold, proceed.

Stage 2 (3–4 weeks): Full experiment matrix (C.3), including collusion and the abelianization-gap control. Prove Theorem 3 (consistency radius) analytically — it is low-risk and publishable regardless. Attempt Theorem 1 with the β_1 -quotient handled explicitly.

Stage 3 (commit gate): Attempt Theorem 2 (localization) and the direct head-to-head vs forensic BFT under partition, targeting the Sheng et al. impossibility boundary.

Commit the chapter iff the prototype demonstrates cohomology-only detection/localization in the partition regime AND at least Theorem 1 or Theorem 2 is provable under stated conditions.

Thresholds that change the recommendation:

- **COMMIT (full chapter)** if: (i) partition experiments show ≥ 1 class of equivocations detected cohomologically but not by pairwise comparison, AND (ii) Theorem 1 or 2 holds. Frame as “cohomology as triage/localization layer feeding forensic proof extraction.”
- **COMMIT (reduced, spectral/consistency-radius section only)** if: consistency radius (Theorem 3) is clean and useful but H^1 adds nothing beyond pairwise — publish the metric, drop the cohomology claims.
- **CUT (appendix remark)** if: the falsifying outcome (C.5) obtains.

Caveats

- **The abelianization gap is real and load-bearing.** Carù (2017/2018) proved strongly contextual (= genuinely inconsistent) models with vanishing Čech obstruction, disproving AMB Conjecture 8.1. Over $\mathbb{R}$ -relaxed coefficients, $H^1 = 0$ does *not* certify global consistency for set-valued log states; treat non-vanishing H^1 as a *sufficient* alarm, never treat vanishing as an all-clear without a separate pairwise sweep.
- **Cyclic-graph contamination:** on any gossip graph with cycles the constant sheaf already has $H^1 = \mathbb{R}^{\beta_1}$; equivocation cohomology must be measured relative to the honest baseline, or the counting theorem is false as naively stated.
- **Cohomology does not produce cryptographic culpability proofs.** It localizes; forensic BFT (Sheng et al.) attributes with non-repudiable evidence. Complementary, not competing; overclaiming equivalence would be an error.
- **Computational tractability is confirmed and *not* threatened by Bach's "#P-hardness":** Bach's result ("Sheaf Cohomology is #P-hard," *J. Symbolic Comput.* 27(4):429–433, 1999) concerns coherent sheaves on projective space specified by graded matrices [ScienceDirect](#) (reducible to lattice-point counting), not finite cellular sheaves over $\mathbb{R}$, which are polynomial-time via linear algebra (rank/kernel of δ , or the Laplacian spectrum). State this explicitly to preempt a reviewer's objection.
- **Ghrist–Krishnan MFMC has known duality gaps** (tightened later by Krishnan and by Krishnamoorthy et al., arXiv:2107.04252); cite the min-cut/repair connection carefully, not as a clean theorem.
- **Some 2025–2026 "sheaf + blockchain/cohomology" sources are of uneven rigor** (e.g., the Scifiniti fragment-sequencing paper, "Blockchain Cohomology"); use as motivation, not as load-bearing citations. The load-bearing literature is Abramsky et al., Hansen–Ghrist, Curry, Robinson, and Sheng et al.
- **PySheaf cannot compute cohomology** (maintainers' own note, quoted above); the cohomology track must be hand-rolled, which is the main (modest) engineering cost.